AGI ALPHA FLYWHEEL

— When Capability Creates Opportunity

Vincent Boucher — President of MONTREAL.AI

27 October 2026 · Research edition

From a private question to a fresh successor mission

This walkthrough follows one offering throughout: a confidential pre-emergency reserve-pool feasibility assessment. The technical construction is executed. Customer values and dollar amounts are explicit modeled scenarios; no transaction occurred.

1. The opportunity

Three hypothetical owners are considering a shared reserve arrangement across six demand locations. They want to decide whether to pay for detailed engineering and contracting. They agree to release one collective feasibility answer but do not authorize a coordinator to collect all raw inventory constraints. A sponsor may pay for the common assessment; each owner keeps its authority over any later asset commitment.

The search did not begin with this application. It combined a documented reserve-coordination need with the ability to compute a coupled function across private inputs. Existing sharing programs and secure computation are acknowledged prior art. The application earns an experiment, not a world-first or market-validation claim.

2. Preserve the useful question

For the original mission, the public family contains every single-demand incident and every pair: 6 + 15 = 21. Each scenario must receive an injective allocation: every simultaneous demand gets a different eligible usable asset. An asset withheld by its owner remains unavailable. The output combines all scenarios.

The structured case `two_universal_assets_pairs_only` has one usable asset at owner 1 and one at owner 2, each eligible for all six demands. The other four padded slots are empty. This invented eligibility is a software test, not an assertion that real transformers are universally interchangeable. Owner 3 still participates in the joint computation and authorized demand question.

Question	Required assets at once	Supplied usable assets	Correct aggregate answer
Every single or pair among six demands	At most 2	2	True
Every single, pair or triple among six demands	At most 3	2	False

This deliberately simple case makes the transformation understandable. Neither raw inventory nor a particular matching has to be published to obtain the agreed answer. It also makes the successor boundary unavoidable: a proof about pairs does not certify triples.

3. Construct and execute

Each owner supplies a fixed-shape private file containing eligibility, scenario availability and withholding bits. Three separate MPyC 0.11 processes secret-share the inputs. For every nonempty demand subset within each scenario, the circuit checks whether there are enough distinct usable neighboring assets. It opens only the conjunction over the complete public family.

The independent cleartext oracle enumerates assignments from the public synthetic fixtures. It does not call the Hall-condition implementation. A bilateral diagnostic, in contrast, can incorrectly approve a pool where every individual demand has a match but simultaneous demands compete for one asset. The independent challenge set includes that contention, owner withholding and correlated owner loss.

The actual three-process challenge returned **true for the 21-scenario pair family and false for the 41-scenario family** in this two-asset case. All sixteen manual challenge executions matched independently specified expected answers and the separate oracle. Three malformed-input controls refused without releasing a verdict. The complete public synthetic fixtures and actual process records are preserved, so this result can be reproduced.

The worker’s local file-read guard and transcript counters provide inspectable execution evidence. The research supervisor nevertheless possesses all synthetic inputs. These processes are not separately operated utilities; the prototype does not establish production host isolation or malicious-security protection.

4. Accept the assessment—and price that assessment

The contract accepts correct complete computation of the declared question. A false result can prevent premature spending; it is not failed delivery. An incomplete scenario family, incorrect computation or missing permission fails acceptance and cannot be turned into a billable negative result.

A transparent central value illustration uses $60,000 of otherwise exposed next-stage planning expenditure, a 35% probability that the answer changes that funding decision, 36 coordination hours avoided at $105.117781 per hour, and a 65% confidence/actionability factor:

Expected pool value = 0.65 × (0.35 × $60,000 + 36 × $105.117781) = approximately $16,110.

Three owners each contribute eight preparation hours at that rate and $100 of endpoint cost:

Total owner participation expense = 3 × (8 × $105.117781 + $100) = approximately $2,823.

Subtracting this input expense and a $12,000 fee leaves $1,286.93 before the remaining participation adjustments. The frozen buyer rule also applies an accepted-delivery belief of 0.8262, calculated as $0.85 \times [1 - 0.08 \times (1 - 0.65)]$, a central privacy burden of $150 per owner, and the opportunity cost of the selected full deposit: $12,000 × 12% × 28/365 = $110.47. Using unrounded inputs:

Complete modeled coalition utility = 0.8262 × ($16,109.7561 − $12,000) − $2,822.8267 − $450 − $110.4658 ≈ $12.19.

This central case only narrowly passes the aggregate financial condition. Consent, budget and every member's participation constraint must also pass. A higher fee cannot retain this central buyer by assumption; participation must be recomputed. If a trusted broker may receive the inputs, its lower cost can defeat this offering. If the answer cannot affect the decision, the assumed information value disappears.

These numbers are neither a quote nor observed willingness to pay. They connect the actual delivered bit to a decision mechanism without monetizing a hypothetical outage or treating an expensive asset as service revenue.

5. Follow complete provider economics

The central repeat-assessment cost is $1,200 before retries and payment effects: orchestration, execution, verification, packaging, recovery/support, settlement/archive and an explicit unknown-cost allowance. The payment fee, noncollection, failed jobs, acquisition, fixed business expenses, insurance allowance and one-time readiness work are separate. The brief MPC runtime is only a component of this budget.

The full paired model finds $2,676 of variable cost per accepted-equivalent at the $12,000 fee, including failed bookings. The resulting $9,324 contribution does not cover its sparse pipeline and complete fixed costs: mean 180-day venture profit is −$30,147. The tested forward decision is no launch.

For a fully inspectable example, the preserved ledger uses seed **9811007**, the first qualifying illustrative accepted-delivery record selected by the study's example rule. It is one modeled world, not an actual customer:

Ledger item	Modeled amount
Qualified prospects / accepted assessments	9 / 1
Accepted service revenue	$12,000.00
Delivery including support / payment processing	$1,452.23 / $57.48
Service contribution	$10,490.29
Startup readiness	$15,214.55
Acquisition / recurring business / insurance	$5,063.71 / $4,625.27 / $2,933.24
Total cost / operating profit	$29,346.48 / −$17,346.48
Day-180 cash, including remaining seed	$42,928.82
Support payable / operating reserve	$275.30 / $4,000.00
Day-180 available funds	$38,653.52
Cash after scheduled support settlement	$42,653.52

The arithmetic keeps a successful assessment and a loss-making venture visible at once. Conditional commitments reduce premature startup spending, but every policy in the separately frozen follow-up still incurred screening expense without assembling sufficient timely customers. Four high-value $18,000 commitments pass the controlled admission branch; that favorable supplied case is a validation target, not a sale.

Deposits are refundable liabilities until acceptance. Seed financing is company capital, not revenue. Customer preparation expense and any physical procurement funds are not provider receipts. Cash reserved for remaining delivery, support, refunds and settlement cannot finance optional learning. The actual research has collected $0.

6. Preserve the opportunity and its evidence

The Nova project contains the actual opportunity, sources, uncertainty, rights and transformation witness. The existing Nova-Seed functions seal and reopen its complete public research genome. A FusionPlan commits the construction, confirmation and bounded next validation specifications. The existing Insight evaluator’s schema refusal is recorded rather than replaced with a fabricated discovery success. MARK checks are local and unfunded; no venture asset was minted or sold.

The native trust and signed-Journal wrapper requires an explicit query grant tied to source, scenario family, input commitments, scope, epoch and output count. It consumes both the nonce and the epoch before execution. Changing a scenario or issuing a new nonce for the same retained slot does not authorize another output. These are local fixture identities with ephemeral signing keys, not live ENS ownership or real utility consent.

7. Fresh successor work

The next mission asks about all triples as well as singles and pairs: 41 scenarios. The successor reuses the unchanged executable, checks the new mission’s meaning and receives a new local grant. Its new generated inputs are independent of the original confirmation inputs. In the worked two-asset challenge, the honest answer changes from true to false because no third distinct asset exists. The useful obligation has become harder; the instrument does not exclude that case or relax it.

A separate recipient context generated and executed 120 fresh successor worlds from seed 810270, without reading the original answers. The instrument matched the oracle on all 120: 84 feasible and 36 not certified; the bilateral diagnostic falsely approved five. Its new grant was consumed before computation, and all 36 source freeze bindings still matched. The ordinary identical-library replay returned exactly the same results.

The original fresh cohort separately matched 120/120, with 85 feasible and 35 not certified. These independent cohorts establish a successful scoped transfer; their positive fractions do not estimate a learning gain.

An ordinary library containing identical information receives the same fresh work. Its equality is informative: portable code can be valuable without an incremental retained-Alpha gain or a graph-specific moat. The baseline already paid to build its product, so ordinary repeat use is not charged a second learning bill. Additional retention or Gym work requires its own justified investment case.

8. The opportunity after this one

The next useful question could include owner minimum-reserve quotas, non-unit capacities or certified transport constraints. That is new scope. It needs appropriate source evidence, a new instrument or verified extension, a new disclosure agreement and new commercial validation. The network receives a tested starting point and clear boundaries—not inherited permission, customer secrets or an unsupported guarantee.

REUSE stable structure. RECHECK the new mission's meaning. REGRANT authority separately.

Reproduce this example

Run `src/challenge_checks.py` for the exact hand-constructed pair/triple cases. The evidence file contains all fixtures, independently specified expected answers, oracle results and actual MPC records. The README supplies the exact command, dependency pins and clean-output instructions. Technical evidence is in `evidence/challenge_checks.json`; the prospective 120 + 120 worlds and ordinary-library replays are under `results/`. The commercial study and example ledger use the same one-assessment offering.